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REFERENCE ELECTRODE MANAGEMENT FOR A BATTERY CELL

Abstract

A method for regulating a state of charge of a reference electrode in a battery cell may include determining a reference electrode state of charge (SOC) of the reference electrode. The battery cell includes an anode, a cathode, and the reference electrode. The reference electrode is disposed between the anode and the cathode. The method further may include comparing the reference electrode SOC to at least a predetermined low SOC threshold. The method further may include adjusting the reference electrode SOC in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold or that the reference electrode SOC is greater than or equal to a predetermined high SOC threshold.

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Background/Summary

INTRODUCTION

[0001] The present disclosure relates to systems and methods for battery management for a battery cell, and more particularly, to systems and methods for regulating a state of charge of a reference electrode in a battery cell.

[0002] To increase performance, ease-of-use, and usable life, it is advantageous to estimate a state of charge (SOC) of a battery cell. Generally, the SOC of a battery cell is not a directly measurable quantity because it is dependent on electrochemical processes occurring within the battery cell. Thus, mathematical models are used to estimate the SOC of the battery cell based on directly measurable quantities such as voltages and/or currents. The mathematical models used to estimate the SOC of the battery cell are often highly non-linear and may be affected by measurement error. Thus, it is advantageous to provide a reference electrode within the battery cell to provide a reference voltage for accurate measurement of voltages of components of the battery cell. However, use of the reference electrode to perform measurements may result in depletion of a SOC of the reference electrode, and thus variation in the reference voltage.

[0003] Thus, while battery management systems and methods achieve their intended purpose, there is a need for a new and improved system and method for regulating a state of charge of a reference electrode in a battery cell.

SUMMARY

[0004] According to several aspects, a method for regulating a state of charge of a reference electrode in a battery cell is provided. The method may include determining a reference electrode state of charge (SOC) of the reference electrode. The battery cell includes an anode, a cathode, and the reference electrode. The reference electrode is disposed between the anode and the cathode. The method further may include comparing the reference electrode SOC to at least a predetermined low SOC threshold. The method further may include adjusting the reference electrode SOC in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold or that the reference electrode SOC is greater than or equal to a predetermined high SOC threshold.

[0005] In another aspect of the present disclosure, determining the reference electrode SOC further may include charging the reference electrode and measuring an amount of charge added to the reference electrode. Determining the reference electrode SOC further may include measuring an anode reference voltage while charging the reference electrode. Determining the reference electrode SOC further may include determining the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode.

[0006] In another aspect of the present disclosure, determining the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode further may include calculating a first order reference electrode voltage slope. The first order reference electrode voltage slope is a first derivative of the anode reference voltage while charging the reference electrode with respect to the amount of charge added to the reference electrode. Determining the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode further may include determining the reference electrode SOC based at least in part on the first order reference electrode voltage slope.

[0007] In another aspect of the present disclosure, determining the reference electrode SOC based at least in part on the first order reference electrode voltage slope further may include determining

the reference electrode SOC to be less than or equal to the predetermined low SOC threshold in response to determining that the first order reference electrode voltage slope is greater than or equal to a predetermined first order reference electrode voltage slope low threshold and that the first order reference electrode voltage slope is decreasing. Determining the reference electrode SOC based at least in part on the first order reference electrode voltage slope further may include determining the reference electrode SOC to be greater than or equal to the predetermined high SOC threshold in response to determining that the first order reference electrode voltage slope is greater than or equal to the predetermined first order reference electrode voltage slope high threshold and that the first order reference electrode voltage slope is increasing.

[0008] In another aspect of the present disclosure, adjusting the reference electrode SOC further may include charging the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold. The reference electrode is charged until the first order reference electrode voltage slope is greater than or equal to the predetermined first order reference electrode voltage slope high threshold and the first order reference electrode voltage slope is increasing.

[0009] In another aspect of the present disclosure, determining the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode further may include calculating a second order reference electrode voltage slope. The second order reference electrode voltage slope is a second derivative of the anode reference voltage while charging the reference electrode with respect to the amount of charge added to the reference electrode. Determining the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode further may include determining the reference electrode SOC based at least in part on the second order reference electrode voltage slope.

[0010] In another aspect of the present disclosure, determining the reference electrode SOC based at least in part on the second order reference electrode voltage slope further may include determining the reference electrode SOC to be less than or equal to the predetermined low SOC threshold in response to determining that the second order reference electrode voltage slope is less than or equal to a predetermined second order reference electrode voltage slope low threshold. Determining the reference electrode SOC based at least in part on the second order reference electrode voltage slope further may include determining the reference electrode SOC to be greater than or equal to the predetermined high SOC threshold in response to determining that the second order reference electrode voltage slope is greater than or equal to a predetermined second order reference electrode voltage slope high threshold.

[0011] In another aspect of the present disclosure, adjusting the reference electrode SOC further may include charging the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold. The reference electrode is charged until the second order reference electrode voltage slope is greater than or equal to the predetermined second order reference electrode voltage slope high threshold.

[0012] In another aspect of the present disclosure, determining the reference electrode SOC further may include tracking an elapsed time since a previous charging process of the reference electrode. Determining the reference electrode SOC further may include calculating an amount of lost charge from the reference electrode based at least in part on the elapsed time and a predetermined reference electrode discharge rate of the reference electrode. Determining the reference electrode SOC further may include calculating the reference electrode SOC based at least in part on the amount of lost charge.

[0013] In another aspect of the present disclosure, adjusting the reference electrode SOC further may include charging the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold. The reference electrode is charged until the amount of lost charge is returned to the reference electrode.

[0014] According to several aspects, a system for regulating a state of charge of a reference electrode in a battery cell is provided. The system may include the battery cell including an anode, a cathode, and the reference electrode. The reference electrode is disposed between the anode and the cathode. The system further may include a reference electrode management system in electrical communication with the battery cell, the reference electrode management system including a controller. The controller is programmed to determine a reference electrode state of charge (SOC) of the reference electrode. The controller is further programmed to compare the reference electrode SOC to at least a predetermined low SOC threshold. The controller is further programmed to adjust the reference electrode SOC in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold or that the reference electrode SOC is greater than or equal to a predetermined high SOC threshold.

[0015] In another aspect of the present disclosure, to determine the reference electrode SOC, the controller is further programmed to charge the reference electrode by allowing a current flow between the cathode and the reference electrode. To determine the reference electrode SOC, the controller is further programmed to measure an amount of charge added to the reference electrode while charging the reference electrode. To determine the reference electrode SOC, the controller is further programmed to measure an anode reference voltage while charging the reference electrode. To determine the reference electrode SOC, the controller is further programmed to determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode.

[0016] In another aspect of the present disclosure, to determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to calculate a first order reference electrode voltage slope. The first order reference electrode voltage slope is a first derivative of the anode reference voltage while charging the reference electrode with respect to the amount of charge added to the reference electrode. To determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to determine the reference electrode SOC to be less than or equal to the predetermined low SOC threshold in response to determining that the first order reference electrode voltage slope is greater than or equal to a predetermined first order reference electrode voltage slope low threshold and that the first order reference electrode voltage slope is decreasing. To determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to determine the reference electrode SOC to be greater than or equal to the predetermined high SOC threshold in response to determining that the first order reference electrode voltage slope is greater than or equal to the predetermined first order reference electrode voltage slope high threshold and that the first order reference electrode voltage slope is increasing.

[0017] In another aspect of the present disclosure, to determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to calculate a second order reference electrode voltage slope. The second order reference electrode voltage slope is a second derivative of the anode reference voltage while charging the reference electrode with respect to the amount of charge added to the reference electrode. To determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to determine the reference electrode SOC to be less than or equal to the predetermined low SOC threshold in response to determining that the second order reference electrode voltage slope is less than or equal to a predetermined second order reference electrode voltage slope low threshold. To determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to determine the reference electrode SOC to be greater than or equal to the

predetermined high SOC threshold in response to determining that the second order reference electrode voltage slope is greater than or equal to a predetermined second order reference electrode voltage slope high threshold.

[0018] In another aspect of the present disclosure, to adjust the reference electrode SOC, the controller is further programmed to charge the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold. The reference electrode is charged until at least one of: (i) the first order reference electrode voltage slope is greater than or equal to the predetermined first order reference electrode voltage slope low threshold and the first order reference electrode voltage slope is increasing and (ii) the second order reference electrode voltage slope is greater than or equal to the predetermined second order reference electrode voltage slope high threshold.

[0019] In another aspect of the present disclosure, to determine the reference electrode SOC, the controller is further programmed to track an elapsed time since a previous charging process of the reference electrode. To determine the reference electrode SOC, the controller is further programmed to calculate an amount of lost charge from the reference electrode based at least in part on the elapsed time and a predetermined reference electrode discharge rate of the reference electrode. To determine the reference electrode SOC, the controller is further programmed to calculate the reference electrode SOC based at least in part on the amount of lost charge.

[0020] In another aspect of the present disclosure, to adjust the reference electrode SOC, the controller is further programmed to charge the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold. The reference electrode is charged until the amount of lost charge is returned to the reference electrode.

[0021] According to several aspects, a system for regulating a state of charge of a reference electrode in a battery cell in a vehicle is provided. The system may include the battery cell including an anode, a cathode, and the reference electrode. The reference electrode is disposed between the anode and the cathode. The system further may include a reference electrode management system in electrical communication with the battery cell. The reference electrode management system includes a controller. The controller is programmed to charge the reference electrode by allowing a current flow between the cathode and the reference electrode. The controller is further programmed to measure an amount of charge added to the reference electrode while charging the reference electrode. The controller is further programmed to measure an anode reference voltage while charging the reference electrode. The controller is further programmed to determine a reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode. The controller is further programmed to compare the reference electrode SOC to at least a predetermined low SOC threshold. The controller is further programmed to adjust the reference electrode SOC in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold or that the reference electrode SOC is greater than or equal to a predetermined high SOC threshold.

[0022] In another aspect of the present disclosure, to determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to calculate a second order reference electrode voltage slope. The second order reference electrode voltage slope is a second derivative of the anode reference voltage while charging the reference electrode with respect to the amount of charge added to the reference electrode. To determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to determine the reference electrode SOC to be less than or equal to the predetermined low SOC threshold in response to determining that the second order reference electrode voltage slope is less than or equal to a predetermined second order reference electrode voltage slope low threshold. To determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is

further programmed to determine the reference electrode SOC to be greater than or equal to the predetermined high SOC threshold in response to determining that the second order reference electrode voltage slope is greater than or equal to a predetermined second order reference electrode voltage slope high threshold.

[0023] In another aspect of the present disclosure, to adjust the reference electrode SOC, the controller is further programmed to charge the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold. The reference electrode is charged until the second order reference electrode voltage slope is greater than or equal to the predetermined second order reference electrode voltage slope high threshold.

[0024] Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

[0026] FIG. 1 is a schematic of a system for regulating a state of charge of a reference electrode in a battery cell, according to an exemplary embodiment;

[0027] FIG. 2 is a schematic diagram of the reference electrode, according to an exemplary embodiment;

[0028] FIG. 3 is a schematic diagram of a reference electrode management system, according to an exemplary embodiment;

[0029] FIG. 4 is a flowchart of a method for regulating a state of charge of a reference electrode in a battery cell, according to an exemplary embodiment;

[0030] FIG. 5 is a flowchart of a first method for determining a reference electrode state of charge, according to an exemplary embodiment;

[0031] FIG. 6 is a flowchart of a first method for determining the reference electrode state of charge based on an anode reference voltage and an amount of charge added during charging, according to an exemplary embodiment;

[0032] FIG. 7 is a first exemplary graph showing an exemplary first order reference voltage slope, according to an exemplary embodiment;

[0033] FIG. 8 is a flowchart of a second method for determining the reference electrode state of charge based on the anode reference voltage and the amount of charge added during charging, according to an exemplary embodiment;

[0034] FIG. 9 is a second exemplary graph with an exemplary second order reference voltage slope, according to an exemplary embodiment;

[0035] FIG. 10 is a flowchart of a second method for determining the reference electrode state of charge, according to an exemplary embodiment;

[0036] FIG. 11 is a flowchart of a method for charging the reference electrode, according to an exemplary embodiment;

[0037] FIG. 12 is a flowchart of a method for discharging the reference electrode, according to an exemplary embodiment; and

[0038] FIG. 13 is a schematic diagram of an exemplary vehicle shown with the system of FIG. 1, an electrical load, and a battery management system, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0039] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.

[0040] In aspects of the present disclosure, it is advantageous to accurately and reliably estimate a state of charge of a battery cell. State of charge estimation may depend on measurement of voltages within the battery cell. Thus, battery cells may include a reference electrode to provide a reference voltage for measurement of specific voltage potentials within the battery cell. However, a state of charge of the reference electrode may become depleted during use, degrading performance. Accordingly, the present disclosure provides a new and improved system and method for regulating a state of charge of a reference electrode in a battery cell.

[0041] Referring to FIG. 1, a schematic of a system **10** for regulating a state of charge of a reference electrode in a battery cell is shown. The system **10** generally includes a battery cell **12** and a reference electrode management system **14**.

[0042] The battery cell **12** is used to store electrical energy in the form of chemical energy. In an exemplary embodiment, the battery cell **12** is a Lithium-Ion battery cell (e.g., a Lithium Cobalt Oxide (LiCoO₂) battery cell, a Lithium Manganese Oxide (LiMn₂O₄) battery cell, a Lithium Iron Phosphate (LiFePO₄) battery cell, a Lithium Nickel Cobalt Aluminum Oxide (LiNiCoAlO₂ or NCA) battery cell, a Lithium Nickel Manganese Cobalt Oxide (LiNiMnCoO₂ or NMC) battery cell, a Lithium Titanate (Li₄Ti₅O₁₂) battery cell, and/or the like). It should be understood that the battery cell **12** may utilize other cell chemistries besides Lithium-Ion without departing from the scope of the present disclosure. In an exemplary embodiment, the battery cell **12** includes a cathode **16**, an anode **18**, a reference electrode **20** disposed between the cathode **16** and the anode **18**, and an electrolyte (not shown) in contact with the cathode **16**, the anode **18**, and the reference electrode **20**.

[0043] In a non-limiting example, the cathode **16** is made of a mixed metal oxide of lithium, nickel, manganese, and cobalt. In a non-limiting example, the anode **18** is made of graphite. In a non-limiting example, the electrolyte includes a lithium salt dissolved in a solvent (e.g., Lithium hexafluorophosphate (LiPF₆), Lithium bis(trifluoromethanesulfonyl) imide (LiTFSI), Lithium perchlorate (LiClO₄), and/or the like). The composition of the reference electrode **20** will be discussed in greater detail below.

[0044] In an exemplary embodiment, the cathode **16** is electrically connected to a positive terminal **22a** of the battery cell **12**. The anode **18** is electrically connected to a negative terminal **22b** of the battery cell **12**. The reference electrode **20** is electrically connected to a reference terminal **22c** of the battery cell **12**. The positive terminal **22a**, negative terminal **22b**, and reference terminal **22c** allow the battery cell **12** to be connected to other systems for the purpose of measuring one or more states of the battery cell **12** and/or providing power to an external device, as will be discussed in greater detail below. The reference terminal **22c** provides a reference electrode voltage used for measurement of a cathode reference voltage measured between the positive terminal **22a** and the reference terminal **22c** and/or an anode reference voltage measured between the negative terminal **22b** and the reference terminal **22c**, as will be discussed in greater detail below.

[0045] In the scope of the present disclosure, a state of charge (SOC) generally refers to an amount or concentration of lithium ions intercalated within a lithium ion intercalation material (i.e., a material capable of intercalating lithium ions) relative to a maximum capacity of the lithium ion intercalation material to intercalate lithium ions. A state of charge (SOC) of the battery cell **12** quantifies a present level of charge stored in the battery cell **12** relative to a maximum charge capacity of the battery cell **12**. On a molecular level, the SOC of the battery cell **12** refers to the distribution of lithium ions between the cathode **16** and the anode **18**. More particularly, the SOC of the battery cell **12** quantifies an amount or concentration of lithium ions intercalated within the anode **18** relative to a maximum capacity of the anode **18** to intercalate lithium ions. In a non-limiting example, when the battery cell **12** is fully charged (i.e., the SOC of the battery cell **12** is 100%), the anode **18** is fully intercalated with lithium ions. As the battery cell **12** discharges, lithium ions move from the anode **18** to the cathode **16** through the electrolyte, resulting in a decrease in the concentration of lithium ions in the anode **18** and an increase in the concentration of

lithium ions in the cathode **16**, thus decreasing the SOC of the battery cell **12**.

[0046] Over charging or over discharging the battery cell **12** may damage components of the battery cell **12** such as the cathode **16** and/or the anode **18**, resulting in a reduction of an overall usable life of the battery cell **12**. Therefore, it is advantageous to determine the SOC of the battery cell **12** for the purposes of battery management. Generally, the SOC of the battery cell **12** is not a directly measurable quantity, and instead must be estimated using a mathematical model of the electrochemical processes occurring within the battery cell **12**. In a non-limiting example, the mathematical model is configured to determine or estimate the SOC of the battery cell **12** based at least in part on an open circuit voltage (OCV) of the battery cell **12**. In another non-limiting example, the mathematical model is configured to determine or estimate the SOC of the battery cell **12** based at least in part on the cathode reference voltage measured between the positive terminal **22a** and the reference terminal **22c** and/or the anode reference voltage measured between the negative terminal **22b** and the reference terminal **22c**.

[0047] Referring to FIG. 2, a schematic diagram of the reference electrode **20** is shown. In an exemplary embodiment, the reference electrode **20** includes a separator sheet **30** that holds a reference strip **32** in place. In a non-limiting example, the separator sheet **30** is made of a thin, porous, electrically insulating material which allows the flow of lithium ions while preventing electrical shorts (e.g., polyethylene (PE), polypropylene (PP), and/or the like). The reference strip **32** includes a conductive strip **34** connected to a reference material **36**. In a non-limiting example, the reference strip **32** is a flexible structure made of a metal foil (e.g., copper, aluminum, and/or the like) or a polymer film (e.g., polyimide, polyethylene terephthalate, and/or the like). The conductive strip **34** is used to provide an electrical connection between the reference material **36** and the reference terminal **22c**. In a non-limiting example, the conductive strip **34** is made of a conductive material (e.g., gold, silver, platinum, copper, and/or the like).

[0048] The reference material **36** is an active material which determines the reference electrode voltage of the reference electrode **20**. In a non-limiting example, the reference material **36** is made of a lithium compound which has a relatively stable and relatively reproducible electrochemical potential over a relatively wide range of lithium concentration. For example, the reference material **36** may include lithium iron phosphate (LiFePO_4), lithium titanate ($\text{Li}_4\text{Ti}_5\text{O}_{12}$), lithium cobalt oxide (LiCoO_2), and/or the like). As discussed above, a state of charge (SOC) generally refers to an amount or concentration of lithium ions intercalated within a lithium ion intercalation material (i.e., a material capable of intercalating lithium ions) relative to a maximum capacity of the lithium ion intercalation material to intercalate lithium ions. Therefore, in the scope of the present disclosure, a reference electrode SOC is defined as an amount or concentration of lithium ions intercalated within the reference material **36** relative to a maximum capacity of the reference material **36** to intercalate lithium ions.

[0049] While the reference electrode voltage of the reference electrode **20** is ideally stable over a wide range of lithium concentration (i.e., reference electrode SOC), variations in the reference electrode SOC may result in variations in the reference electrode voltage, decreasing the accuracy of cathode reference voltage and/or the anode reference voltage measurements. In a non-limiting example, variations in the reference electrode SOC may be caused by electrochemical reactions (also referred to as side reactions) occurring within the battery cell **12**. In another non-limiting example, variations in the reference electrode SOC may be caused by charge and/or discharge of the reference electrode material induced by external measurement circuitry in the process of conducting measurements of the cathode reference voltage and/or the anode reference voltage. Therefore, it is advantageous to regulate the reference electrode SOC and maintain the reference electrode SOC within a known and/or ideal range, as will be discussed in greater detail below.

[0050] Referring again to FIG. 1, the reference electrode management system **14** is used to regulate the reference electrode SOC. As shown in FIG. 1, the reference electrode management system **14** is in electrical communication with the positive terminal **22a**, the negative terminal **22b**, and the

reference terminal **22c**. In an exemplary embodiment, the reference electrode management system **14** is affixed to the battery cell **12**, such that the reference electrode management system **14** may operate even when the battery cell **12** is not installed in another device or module (e.g., a battery pack). While the reference electrode management system **14** is depicted in FIG. **1** as being affixed to an outer casing of the battery cell **12**, it should be understood that the reference electrode management system **14** may be integrated into the outer casing of the battery cell **12**, located within the battery cell **12** with internal connections to the cathode **16**, anode **18**, and reference electrode **20**, or otherwise integral with the battery cell **12** without departing from the scope of the present disclosure.

[0051] In another exemplary embodiment, the reference electrode management system **14** is a modular component configured to be easily installable, removable, and replaceable on the battery cell **12**. In another exemplary embodiment, the reference electrode management system **14** is located remotely from the battery cell **12** and is electrically connected to the battery cell **12**. In another exemplary embodiment, the reference electrode management system **14** is configured to regulate the reference electrode SOC of a plurality of battery cells as part of a battery system (e.g., a multi-cell battery module/pack) and is electrically connected with the plurality of battery cells.

[0052] Referring to FIG. **3**, a schematic diagram of the reference electrode management system **14** is shown. In an exemplary embodiment, the reference electrode management system **14** includes at least a controller **40** and an interface circuit **42**.

[0053] The controller **40** is used to implement a method **100** for regulating a state of charge of a reference electrode in a battery cell, as will be described below. The controller **40** includes at least one processor **44** and a non-transitory computer readable storage device or media **46**. The processor **44** may be a custom made or commercially available processor, a central processing unit (CPU), a graphics processing unit (GPU), an auxiliary processor among several processors associated with the controller **40**, a semiconductor-based microprocessor (in the form of a microchip or chip set), a macroprocessor, a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a combination thereof, or generally a device for executing instructions.

[0054] The computer readable storage device or media **46** may include volatile and nonvolatile storage in read-only memory (ROM), random-access memory (RAM), and keep-alive memory (KAM), for example. KAM is a persistent or non-volatile memory that may be used to store various operating variables while the processor **44** is powered down. The computer-readable storage device or media **46** may be implemented using a number of memory devices such as PROMs (programmable read-only memory), EPROMs (electrically PROM), EEPROMs (electrically erasable PROM), flash memory, or another electric, magnetic, optical, or combination memory devices capable of storing data, some of which represent executable instructions used by the controller **40** to control the reference electrode management system **14**. The controller **40** may also consist of multiple controllers which are in electrical communication with each other. The controller **40** further may include additional elements and/or modules, such as, for example, a real-time clock (RTC) module for measuring the passage of real-time. In an exemplary embodiment, the controller **40** is powered by connection to the positive terminal **22a** and the negative terminal **22b** of the battery cell **12**.

[0055] The controller **40** is in electrical communication with the interface circuit **42**. In an exemplary embodiment, the electrical communication is established using, for example, general purpose input/output (GPIO) pins, an inter-integrated circuit (I2C) bus, a serial peripheral interface (SPI) bus, a parallel communication bus, or the like. It should be understood that various additional communication protocols for communicating with the controller **40** are within the scope of the present disclosure.

[0056] The interface circuit **42** is used to interface the controller **40** with the positive terminal **22a**, the negative terminal **22b**, and the reference terminal **22c**. In an exemplary embodiment, the interface circuit **42** includes a measurement circuit **48** and a charging circuit **50**.

[0057] The measurement circuit **48** is used to measure the anode reference voltage, the cathode reference voltage, and a current flow into/out of the reference electrode **20**. In a non-limiting example, the measurement circuit **48** includes, for example an analog to digital converter (ADC). The measurement circuit **48** further may include additional components to support voltage measurement, including, for example, a voltage follower, an input buffer, a multiplexer, and/or the like. The measurement circuit **48** further includes components allowing the controller **40** to measure the current flow into/out of the reference electrode **20**, including, for example, a shunt resistor, an electromagnetic current sensor, an ADC, and/or the like.

[0058] The charging circuit **50** is used to charge and/or discharge the reference electrode **20**. In an exemplary embodiment, the charging circuit **50** includes switching electronics allowing the controller **40** to connect and disconnect the reference terminal **22c** between the positive terminal **22a** or the negative terminal **22b** to charge or discharge the reference electrode **20**. In a non-limiting example, the charging circuit **50** includes, for example, relays, contractors, transistors, and/or the like. It should be understood that the measurement circuit **48** and/or the charging circuit **50** of the interface circuit **42** further may include additional passive or active analog and/or digital electronics such as, for example, resistors, capacitors, inductors, filters, amplifiers, power electronics, digital to analog converters (DAC), and/or the like. In an exemplary embodiment, the interface circuit **42** is powered by connection to the positive terminal **22a** and the negative terminal **22b** of the battery cell **12**. The interface circuit **42** is in electrical communication with the controller **40** as discussed above.

[0059] Referring to FIG. **4**, a flowchart of the method **100** for regulating a state of charge of a reference electrode in a battery cell is shown. The method **100** begins at block **102** and proceeds to block **104**. At block **104**, the controller **40** determines the reference electrode SOC of the reference electrode **20**. Methods for determining the reference electrode SOC will be discussed in greater detail below. After block **104**, the method **100** proceeds to block **106**.

[0060] At block **106**, the controller **40** compares the reference electrode SOC determined at block **104** to a predetermined low SOC threshold (e.g., 10%) and a predetermined high SOC threshold (e.g., 90%). If the reference electrode SOC is less than or equal to the predetermined low SOC threshold, the method **100** proceeds to block **108**. If the reference electrode SOC is greater than the predetermined low SOC threshold and less than the predetermined high SOC threshold, the method **100** proceeds to enter a standby state at block **110**. If the reference electrode SOC is greater than or equal to the predetermined high SOC threshold, the method **100** proceeds to block **112**, as will be discussed in greater detail below.

[0061] At block **108**, the controller **40** uses the interface circuit **42** to charge the reference electrode **20** to adjust the reference electrode SOC in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold. Methods for charging the reference electrode **20** will be discussed in greater detail below. After block **108**, the method **100** proceeds to enter the standby state at block **110**.

[0062] At block **112**, the controller **40** uses the interface circuit **42** to discharge the reference electrode **20** to adjust the reference electrode SOC in response to determining that the reference electrode SOC is greater than or equal to the predetermined high SOC threshold. Methods for discharging the reference electrode **20** will be discussed in greater detail below. After block **112**, the method **100** proceeds to enter the standby state at block **110**.

[0063] Referring to FIG. **5**, a flowchart of a first exemplary embodiment **104a** of block **104** of the method **100** (i.e., a first method for determining the reference electrode SOC) is shown. The first exemplary embodiment **104a** begins at blocks **502**, **504**, and **506**. At block **502**, the controller **40** uses the interface circuit **42** to charge the reference electrode **20**. In an exemplary embodiment, to charge the reference electrode **20**, the controller **40** uses the charging circuit **50** to connect the positive terminal **22a** to the reference terminal **22c** allowing flow of current from the cathode **16** to the reference electrode **20**. In an exemplary embodiment, the charging circuit **50** allows current

flow for a predetermined duration. In a non-limiting example, the predetermined duration depends at least in part on a cross-sectional area and/or a volume of the reference electrode **20**. In another exemplary embodiment, the charging circuit **50** allows current to flow until the reference electrode SOC has increased by a predetermined amount (e.g., five percent). After block **502**, the first exemplary embodiment **104a** proceeds to block **508**, as will be discussed in greater detail below. [0064] At block **504**, the controller **40** uses the measurement circuit **48** to measure the current flow into the reference electrode **20** during charging at block **502**. Based on the predetermined duration, the controller **40** determines an amount of charge added to the reference electrode **20** during charging at block **502**. After block **504**, the first exemplary embodiment **104a** proceeds to block **508**, as will be discussed in greater detail below.

[0065] At block **506**, the controller **40** uses the measurement circuit **48** to measure the anode reference voltage during charging at block **502**. After block **506**, the first exemplary embodiment **104a** proceeds to block **508**.

[0066] At block **508**, the controller **40** determines the reference electrode SOC based at least in part on the anode reference voltage during charging and the amount of charge added to the reference electrode **20** during charging. Methods for determining the reference electrode SOC based at least in part on the anode reference voltage during charging and the amount of charge added to the reference electrode **20** during charging are discussed in greater detail below. After block **508**, the first exemplary embodiment **104a** is concluded, and the method **100** proceeds as discussed above.

[0067] Referring to FIG. **6**, a flowchart of a first exemplary embodiment **508a** of block **508** of the first exemplary embodiment **104a** of block **104** of the method **100** (i.e., a first method for determining the reference electrode SOC based on anode reference voltage and amount of charge added during charging) is shown. The first exemplary embodiment **508a** begins at block **602**. At block **602**, the controller **40** calculates a first order reference electrode voltage slope. In the scope of the present disclosure, the first order reference electrode voltage slope is a first derivative of the anode reference voltage while charging the reference electrode **20** (as determined at block **506**) with respect to the amount of charge added to the reference electrode **20** (as determined at block **508**).

[0068] Referring to FIG. **7**, a first exemplary graph **70** with an exemplary first order reference voltage slope **72** determined over a range of reference electrode SOC is shown. An x-axis **74** of the first exemplary graph **70** represents the reference electrode SOC. A y-axis **76** of the first exemplary graph **70** represents a value of the first order reference voltage slope. A first dashed line of the first exemplary graph **70** represents a predetermined first order reference electrode voltage slope low threshold **78a**. A second dashed line of the first exemplary graph **70** represents a predetermined first order reference electrode voltage slope high threshold **78b**. In an exemplary embodiment, the predetermined first order reference electrode voltage slope low and high thresholds **78a**, **78b** are chosen based at least in part on the predetermined low SOC threshold and/or the predetermined high SOC threshold.

[0069] It should be understood that the exemplary first order reference voltage slope **72** is exemplary in nature and that the shape and values of the first order reference electrode voltage slope may vary based on, for example, application specific characteristics of the reference electrode **20** and/or the battery cell **12**. It should further be understood that the value of the predetermined first order reference electrode voltage slope low and high thresholds **78a**, **78b** shown in the first exemplary graph **70** are merely exemplary in nature. In some embodiments, the predetermined first order reference electrode voltage slope low and high thresholds **78a**, **78b** may be equal in value. Furthermore, the first exemplary graph **70**, the predetermined first order reference electrode voltage slope low and high thresholds **78a**, **78b**, and the exemplary first order reference voltage slope **72** are not necessarily to scale.

[0070] Referring again to FIG. **6** and with continued reference to FIG. **7**, after block **602**, the first exemplary embodiment **508a** proceeds to block **604**. At block **604**, the controller **40** compares the

first order reference electrode voltage slope determined at block **602** to the predetermined first order reference electrode voltage slope low and high thresholds **78a**, **78b**. If the first order reference electrode voltage slope determined at block **602** is greater than or equal to the predetermined first order reference electrode voltage slope low threshold **78a** and decreasing as charge is added to the reference electrode **20**, the first exemplary embodiment **508a** proceeds to block **606**. If the first order reference electrode voltage slope determined at block **602** is greater than or equal to the predetermined first order reference electrode voltage slope high threshold **78b** and increasing as charge is added to the reference electrode **20**, the first exemplary embodiment **508a** proceeds to block **608**, as will be discussed in greater detail below.

[0071] At block **606**, the reference electrode SOC is determined to be less than or equal to the predetermined low SOC threshold in response to determining that the first order reference electrode voltage slope determined at block **602** is greater than or equal to the predetermined first order reference electrode voltage slope low threshold **78a** and decreasing as charge is added to the reference electrode **20**. After block **606**, the first exemplary embodiment **508a** is concluded, and the first exemplary embodiment **104a** proceeds as discussed above.

[0072] At block **608**, the reference electrode SOC is determined to be greater than or equal to the predetermined high SOC threshold in response to determining that the first order reference electrode voltage slope determined at block **602** is greater than or equal to the predetermined first order reference electrode voltage slope high threshold **78b** and increasing as charge is added to the reference electrode **20**. After block **608**, the first exemplary embodiment **508a** is concluded, and the first exemplary embodiment **104a** proceeds as discussed above.

[0073] Referring to FIG. **8**, a flowchart of a second exemplary embodiment **508b** of block **508** of the first exemplary embodiment **104a** of block **104** of the method **100** (i.e., a second method for determining the reference electrode SOC based on anode reference voltage and amount of charge added during charging) is shown. It should be understood that the first exemplary embodiment **508a**, the second exemplary embodiment **508b**, or any combination thereof may be used to perform block **508** of the first exemplary embodiment **104a** in the scope of the present disclosure. The second exemplary embodiment **508b** begins at block **802**. At block **802**, the controller **40** calculates a second order reference electrode voltage slope. In the scope of the present disclosure, the second order reference electrode voltage slope is a second derivative of the anode reference voltage while charging the reference electrode **20** (as determined at block **506**) with respect to the amount of charge added to the reference electrode **20** (as determined at block **508**).

[0074] Referring to FIG. **9**, a second exemplary graph **80** with an exemplary second order reference voltage slope **82** determined over a range of reference electrode SOC is shown. An x-axis **84** of the second exemplary graph **80** represents the reference electrode SOC. A y-axis **86** of the second exemplary graph **80** represents a value of the second order reference voltage slope. A first dashed line of the second exemplary graph **80** represents a predetermined second order reference electrode voltage slope low threshold **88a**. A second dashed line of the second exemplary graph **80** represents a predetermined second order reference electrode voltage slope high threshold **88b**. In an exemplary embodiment, the predetermined second order reference electrode voltage slope low and high thresholds **88a**, **88b** are chosen based at least in part on the predetermined low SOC threshold and/or the predetermined high SOC threshold.

[0075] It should be understood that the exemplary second order reference voltage slope **82** is exemplary in nature and that the shape and values of the second order reference electrode voltage slope may vary based on, for example, application specific characteristics of the reference electrode **20** and/or the battery cell **12**. It should further be understood that the value of the predetermined second order reference electrode voltage slope low and high thresholds **88a**, **88b** shown in the second exemplary graph **80** are merely exemplary in nature. In some embodiments, the predetermined second order reference electrode voltage slope low and high thresholds **88a**, **88b** may be equal in magnitude. Furthermore, the second exemplary graph **80**, the predetermined

second order reference electrode voltage slope low and high thresholds **88a**, **88b**, and the exemplary second order reference voltage slope **82** are not necessarily to scale.

[0076] Referring again to FIG. **8** and with continued reference to FIG. **9**, after block **802**, the second exemplary embodiment **508b** proceeds to block **804**. At block **804**, the controller **40** compares the second order reference electrode voltage slope determined at block **802** to the predetermined second order reference electrode voltage slope low and high thresholds **88a**, **88b**. If the second order reference electrode voltage slope determined at block **802** is less than or equal to the predetermined second order reference electrode voltage slope low threshold **88a**, the second exemplary embodiment **508b** proceeds to block **806**. If the second order reference electrode voltage slope determined at block **802** is greater than or equal to the predetermined second order reference electrode voltage slope high threshold **88b**, the second exemplary embodiment **508b** proceeds to block **808**, as will be discussed in greater detail below.

[0077] At block **806**, the reference electrode SOC is determined to be less than or equal to the predetermined low SOC threshold in response to determining that the second order reference electrode voltage slope determined at block **802** is less than or equal to the predetermined second order reference electrode voltage slope low threshold **88a**. After block **806**, the second exemplary embodiment **508b** is concluded, and the first exemplary embodiment **104a** proceeds as discussed above.

[0078] At block **808**, the reference electrode SOC is determined to be greater than or equal to the predetermined high SOC threshold in response to determining that the second order reference electrode voltage slope determined at block **802** is greater than or equal to the predetermined second order reference electrode voltage slope high threshold **88b**. After block **808**, the second exemplary embodiment **508b** is concluded, and the first exemplary embodiment **104a** proceeds as discussed above.

[0079] Referring to FIG. **10**, a flowchart of a second exemplary embodiment **104b** of block **104** of the method **100** (i.e., a second method for determining the reference electrode SOC) is shown. It should be understood that the first exemplary embodiment **104a**, the second exemplary embodiment **104b**, or any combination thereof may be used to perform block **104** of the method **100** in the scope of the present disclosure. The second exemplary embodiment **104b** begins at block **1002**. At block **1002**, the controller **40** tracks an elapsed time since the reference electrode **20** was previously charged (i.e., an elapsed time since a previous execution of block **108** of the method **100**). In an exemplary embodiment, the controller **40** uses a real-time clock (RTC) module to track the elapsed time. After block **1002**, the second exemplary embodiment **104b** proceeds to block **1004**.

[0080] At block **1004**, the controller **40** calculates the reference electrode SOC. In an exemplary embodiment, the controller **40** first calculates an amount of lost charge lost from the reference electrode **20** over the elapsed time determined at block **1002**. In a non-limiting example, the controller **40** multiplies the elapsed time by a predetermined reference electrode discharge rate of the reference electrode. In the scope of the present disclosure, the predetermined reference electrode discharge rate quantifies an amount of charge lost from the reference electrode **20** per unit of time during normal operation of the reference electrode management system **14**. In an exemplary embodiment, the predetermined reference electrode discharge rate is determined using computer simulation and/or physical experimentation. The predetermined reference electrode discharge rate is then saved in the media **46** of the controller **40** for use during the second exemplary embodiment **104b**.

[0081] The controller **40** then calculates the reference electrode SOC based on the amount of lost charge. In an exemplary embodiment, the reference electrode SOC is proportional to a previous amount of charge on the reference electrode **20** (as determined, for example, based on the previous charging process of the reference electrode **20** using the measurement circuit **48** of the interface circuit **42**) minus the amount of lost charge. After block **1004**, the second exemplary embodiment

104b is concluded, and the method **100** proceeds as discussed above.

[0082] Referring to FIG. **11**, a flowchart of an exemplary embodiment **108a** of block **108** of the method **100** (i.e., a method for charging the reference electrode **20**) is shown. The exemplary embodiment **108a** begins at block **1102**. At block **1102**, the controller **40** charges the reference electrode **20**. In an exemplary embodiment, to charge the reference electrode **20**, the controller **40** uses the charging circuit **50** of the interface circuit **42** to connect the positive terminal **22a** (i.e., the cathode **16**) to the reference terminal **22c** (i.e., the reference electrode **20**), allowing current to flow into the reference electrode **20**. In an exemplary embodiment, the controller **40** concurrently uses the measurement circuit **48** to measure a magnitude of the current flow into the reference electrode **20** over time for purposes of calculation of charge added to the reference electrode **20**. The controller **40** also uses the measurement circuit **48** to measure the anode reference voltage while charging the reference electrode **20**. After block **1102**, the exemplary embodiment **108a** proceeds to block **1104**.

[0083] At block **1104**, the controller **40** evaluates a charging stopping condition. In a first exemplary embodiment, the charging stopping condition is satisfied when the first order reference electrode voltage slope (as determined using the method described in reference to the first exemplary embodiment **508a** of block **508** of the first exemplary embodiment **104a** of block **104** of the method **100**) is greater than or equal to the predetermined first order reference electrode voltage slope high threshold **78b** and the first order reference electrode voltage slope is increasing. In other words, the charging stopping condition is satisfied when the reference electrode SOC reaches the predetermined high SOC threshold.

[0084] In a second exemplary embodiment, the charging stopping condition is satisfied when the second order reference electrode voltage slope (as determined using the method described in reference to the second exemplary embodiment **508b** of block **508** of the first exemplary embodiment **104a** of block **104** of the method **100**) is greater than or equal to the predetermined second order reference electrode voltage slope high threshold **88b**. In other words, the charging stopping condition is satisfied when the reference electrode SOC reaches the predetermined high SOC threshold.

[0085] In a third exemplary embodiment, the charging stopping condition is satisfied when the amount of lost charge (as determined using the method described in reference to the second exemplary embodiment **104b** of block **104** of the method **100**) is returned to the reference electrode **20**. If the charging stopping condition is not satisfied, the exemplary embodiment **108a** returns to block **1102** to continue charging the reference electrode **20**. If the charging stopping condition is satisfied, the exemplary embodiment **108a** is concluded, and the method **100** proceeds as described above.

[0086] Referring to FIG. **12**, a flowchart of an exemplary embodiment **112a** of block **112** of the method **100** (i.e., a method for discharging the reference electrode **20**) is shown. The exemplary embodiment **112a** begins at block **1202**. At block **1202**, the controller **40** discharges the reference electrode **20**. In an exemplary embodiment, to discharge the reference electrode **20**, the controller **40** uses the charging circuit **50** of the interface circuit **42** to connect the negative terminal **22b** (i.e., the anode **18**) to the reference terminal **22c** (i.e., the reference electrode **20**), allowing current to flow out of the reference electrode **20**. In an exemplary embodiment, the controller **40** concurrently uses the measurement circuit **48** to measure a magnitude of the current flow out of the reference electrode **20** over time for purposes of calculation of charge removed from the reference electrode **20**. The controller **40** also uses the measurement circuit **48** to measure the anode reference voltage while discharging the reference electrode **20**. After block **1202**, the exemplary embodiment **112a** proceeds to block **1204**.

[0087] At block **1204**, the controller **40** evaluates a discharging stopping condition. In a first exemplary embodiment, the discharging stopping condition is satisfied when the first order reference electrode voltage slope (as determined using the method described in reference to the first

exemplary embodiment **508a** of block **508** of the first exemplary embodiment **104a** of block **104** of the method **100**) is greater than or equal to the predetermined first order reference electrode voltage slope high threshold **78b** and the first order reference electrode voltage slope is increasing. In other words, the discharging stopping condition is satisfied when the reference electrode SOC reaches the predetermined high SOC threshold.

[0088] In a second exemplary embodiment, the discharging stopping condition is satisfied when the second order reference electrode voltage slope (as determined using the method described in reference to the second exemplary embodiment **508b** of block **508** of the first exemplary embodiment **104a** of block **104** of the method **100**) is greater than or equal to the predetermined second order reference electrode voltage slope high threshold **88b**. In other words, the discharging stopping condition is satisfied when the reference electrode SOC reaches the predetermined high SOC threshold. If the discharging stopping condition is satisfied, the exemplary embodiment **112a** is concluded, and the method **100** proceeds as described above.

[0089] Referring to FIG. **13**, an exemplary vehicle **90** is shown with the system **10**, an electrical load **92**, and a battery management system **94**. The electrical load **92** and the battery management system **94** are in electrical communication with the system **10**. In an exemplary embodiment, the electrical load **92** includes electrical and/or electromechanical components or systems of the vehicle **90** which require electrical energy for operation. In a non-limiting example, the electrical load **92** includes the battery management system **94**, a vehicle controller, an infotainment system, one or more vehicle lights, a vehicle propulsion system including an electric motor, and/or the like. The battery cell **12** of the system **10** is configured to provide electrical energy to the electrical load **92**.

[0090] The battery management system **94** is used to monitor the battery cell **12**, provide information about a state of the battery cell **12** to external systems (e.g., the vehicle controller), and optimize use of the battery cell **12** to prolong a usable life of the battery cell **12** and protect the battery cell **12** from damage. In an exemplary embodiment, the battery management system **94** facilitates the electrical connection between the system **10** and the electrical load **92** and may disconnect the system **10** from the electrical load **92** for protection of the battery cell **12**. In an exemplary embodiment, the battery management system **94** performs measurements of the anode reference voltage and/or the cathode reference voltage (using, for example, the measurement circuit **48** of the interface circuit **42** of the reference electrode management system **14**) and executes a mathematical model to estimate the SOC of the battery cell **12**. The SOC of the battery cell **12** is used by other vehicle systems (e.g., the vehicle controller) to provide vehicle range and/or battery status information to an occupant of the vehicle **90**.

[0091] The system **10** and method **100** of the present disclosure offer several advantages. Using the first exemplary embodiment **104a** of block **104** of the method **100** (i.e., the first method for determining the reference electrode SOC), the reference electrode SOC may be determined regardless of the SOC of the battery cell **12**. Therefore, the reference electrode SOC may be monitored and adjusted frequently to prevent over discharging of the reference electrode **20**. Using the second exemplary embodiment **104b** of block **104** of the method **100** (i.e., the second method for determining the reference electrode SOC), the reference electrode SOC may be determined regardless of the SOC of the battery cell **12** and regardless of an equilibrium state of the battery cell **12** (i.e., when the battery cell **12** is under any load condition). Therefore, the reference electrode SOC may be monitored and adjusted frequently to prevent over discharging of the reference electrode **20**, even when the battery cell **12** is under frequent load. In an exemplary embodiment, the first exemplary embodiment **104a** and the second exemplary embodiment **104b** are used in combination to provide a balance between accuracy and frequency of measurement of the reference electrode SOC. Furthermore, by providing the reference electrode management system **14** affixed to or otherwise integral with the battery cell **12**, the reference electrode management system **14** may operate to regulate the reference electrode SOC even if the battery cell **12** is not installed in a

host application (e.g., if the battery cell **12** is in storage or transport), prolonging a usable life of the reference electrode **20** and the battery cell **12**.

[0092] The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

Claims

1. A method for regulating a state of charge of a reference electrode in a battery cell, the method comprising: determining a reference electrode state of charge (SOC) of the reference electrode, wherein the battery cell includes an anode, a cathode, and the reference electrode, and wherein the reference electrode is disposed between the anode and the cathode; comparing the reference electrode SOC to at least a predetermined low SOC threshold; and adjusting the reference electrode SOC in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold or that the reference electrode SOC is greater than or equal to a predetermined high SOC threshold.

2. The method of claim 1, wherein determining the reference electrode SOC further comprises: charging the reference electrode and measuring an amount of charge added to the reference electrode; measuring an anode reference voltage while charging the reference electrode; and determining the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode.

3. The method of claim 2, wherein determining the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode further comprises: calculating a first order reference electrode voltage slope, wherein the first order reference electrode voltage slope is a first derivative of the anode reference voltage while charging the reference electrode with respect to the amount of charge added to the reference electrode; and determining the reference electrode SOC based at least in part on the first order reference electrode voltage slope.

4. The method of claim 3, wherein determining the reference electrode SOC based at least in part on the first order reference electrode voltage slope further comprises: determining the reference electrode SOC to be less than or equal to the predetermined low SOC threshold in response to determining that the first order reference electrode voltage slope is greater than or equal to a predetermined first order reference electrode voltage slope low threshold and that the first order reference electrode voltage slope is decreasing; and determining the reference electrode SOC to be greater than or equal to the predetermined high SOC threshold in response to determining that the first order reference electrode voltage slope is greater than or equal to the predetermined first order reference electrode voltage slope high threshold and that the first order reference electrode voltage slope is increasing.

5. The method of claim 4, wherein adjusting the reference electrode SOC further comprises: charging the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold, wherein the reference electrode is charged until the first order reference electrode voltage slope is greater than or equal to the predetermined first order reference electrode voltage slope high threshold and the first order reference electrode voltage slope is increasing.

6. The method of claim 2, wherein determining the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode further comprises: calculating a second order reference electrode voltage slope, wherein the second order reference electrode voltage slope is a second derivative of the anode reference voltage while charging the reference electrode with respect to the amount of charge added to the reference

electrode; and determining the reference electrode SOC based at least in part on the second order reference electrode voltage slope.

7. The method of claim 6, wherein determining the reference electrode SOC based at least in part on the second order reference electrode voltage slope further comprises: determining the reference electrode SOC to be less than or equal to the predetermined low SOC threshold in response to determining that the second order reference electrode voltage slope is less than or equal to a predetermined second order reference electrode voltage slope low threshold; and determining the reference electrode SOC to be greater than or equal to the predetermined high SOC threshold in response to determining that the second order reference electrode voltage slope is greater than or equal to a predetermined second order reference electrode voltage slope high threshold.

8. The method of claim 7, wherein adjusting the reference electrode SOC further comprises: charging the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold, wherein the reference electrode is charged until the second order reference electrode voltage slope is greater than or equal to the predetermined second order reference electrode voltage slope high threshold.

9. The method of claim 1, wherein determining the reference electrode SOC further comprises: tracking an elapsed time since a previous charging process of the reference electrode; calculating an amount of lost charge from the reference electrode based at least in part on the elapsed time and a predetermined reference electrode discharge rate of the reference electrode; and calculating the reference electrode SOC based at least in part on the amount of lost charge.

10. The method of claim 9, wherein adjusting the reference electrode SOC further comprises: charging the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold, wherein the reference electrode is charged until the amount of lost charge is returned to the reference electrode.

11. A system for regulating a state of charge of a reference electrode in a battery cell, the system comprising: the battery cell including an anode, a cathode, and the reference electrode, wherein the reference electrode is disposed between the anode and the cathode; a reference electrode management system in electrical communication with the battery cell, the reference electrode management system including a controller, wherein the controller is programmed to: determine a reference electrode state of charge (SOC) of the reference electrode; compare the reference electrode SOC to at least a predetermined low SOC threshold; and adjust the reference electrode SOC in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold or that the reference electrode SOC is greater than or equal to a predetermined high SOC threshold.

12. The system of claim 11, wherein to determine the reference electrode SOC, the controller is further programmed to: charge the reference electrode by allowing a current flow between the cathode and the reference electrode; measure an amount of charge added to the reference electrode while charging the reference electrode; measure an anode reference voltage while charging the reference electrode; and determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode.

13. The system of claim 12, wherein to determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to: calculate a first order reference electrode voltage slope, wherein the first order reference electrode voltage slope is a first derivative of the anode reference voltage while charging the reference electrode with respect to the amount of charge added to the reference electrode; determine the reference electrode SOC to be less than or equal to the predetermined low SOC threshold in response to determining that the first order reference electrode voltage slope is greater than or equal to a predetermined first order reference electrode voltage slope low threshold and that the first order reference electrode voltage slope is decreasing; and determine the reference electrode SOC to be greater than or equal to the predetermined high SOC

threshold in response to determining that the first order reference electrode voltage slope is greater than or equal to the predetermined first order reference electrode voltage slope high threshold and that the first order reference electrode voltage slope is increasing.

14. The system of claim 13, wherein to determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to: calculate a second order reference electrode voltage slope, wherein the second order reference electrode voltage slope is a second derivative of the anode reference voltage while charging the reference electrode with respect to the amount of charge added to the reference electrode; and determine the reference electrode SOC to be less than or equal to the predetermined low SOC threshold in response to determining that the second order reference electrode voltage slope is less than or equal to a predetermined second order reference electrode voltage slope low threshold; and determine the reference electrode SOC to be greater than or equal to the predetermined high SOC threshold in response to determining that the second order reference electrode voltage slope is greater than or equal to a predetermined second order reference electrode voltage slope high threshold.

15. The system of claim 14, wherein to adjust the reference electrode SOC, the controller is further programmed to: charge the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold, wherein the reference electrode is charged until at least one of: (i) the first order reference electrode voltage slope is greater than or equal to the predetermined first order reference electrode voltage slope high threshold and the first order reference electrode voltage slope is increasing and (ii) the second order reference electrode voltage slope is greater than or equal to the predetermined second order reference electrode voltage slope high threshold.

16. The system of claim 11, wherein to determine the reference electrode SOC, the controller is further programmed to: track an elapsed time since a previous charging process of the reference electrode; calculate an amount of lost charge from the reference electrode based at least in part on the elapsed time and a predetermined reference electrode discharge rate of the reference electrode; and calculate the reference electrode SOC based at least in part on the amount of lost charge.

17. The system of claim 16, wherein to adjust the reference electrode SOC, the controller is further programmed to: charge the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold, wherein the reference electrode is charged until the amount of lost charge is returned to the reference electrode.

18. A system for regulating a state of charge of a reference electrode in a battery cell in a vehicle, the system comprising: the battery cell including an anode, a cathode, and the reference electrode, wherein the reference electrode is disposed between the anode and the cathode; a reference electrode management system in electrical communication with the battery cell, the reference electrode management system including a controller, wherein the controller is programmed to: charge the reference electrode by allowing a current flow between the cathode and the reference electrode; measure an amount of charge added to the reference electrode while charging the reference electrode; measure an anode reference voltage while charging the reference electrode; determine a reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode; compare the reference electrode SOC to at least a predetermined low SOC threshold; and adjust the reference electrode SOC in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold or that the reference electrode SOC is greater than or equal to a predetermined high SOC threshold.

19. The system of claim 18, wherein to determine the reference electrode SOC based at least in part on the anode reference voltage and the amount of charge added to the reference electrode, the controller is further programmed to: calculate a second order reference electrode voltage slope, wherein the second order reference electrode voltage slope is a second derivative of the anode

reference voltage while charging the reference electrode with respect to the amount of charge added to the reference electrode; and determine the reference electrode SOC to be less than or equal to the predetermined low SOC threshold in response to determining that the second order reference electrode voltage slope is less than or equal to a predetermined second order reference electrode voltage slope low threshold; and determine the reference electrode SOC to be greater than or equal to the predetermined high SOC threshold in response to determining that the second order reference electrode voltage slope is greater than or equal to a predetermined second order reference electrode voltage slope high threshold.

20. The system of claim 19, wherein to adjust the reference electrode SOC, the controller is further programmed to: charge the reference electrode in response to determining that the reference electrode SOC is less than or equal to the predetermined low SOC threshold, wherein the reference electrode is charged until the second order reference electrode voltage slope is greater than or equal to the predetermined second order reference electrode voltage slope high threshold.
